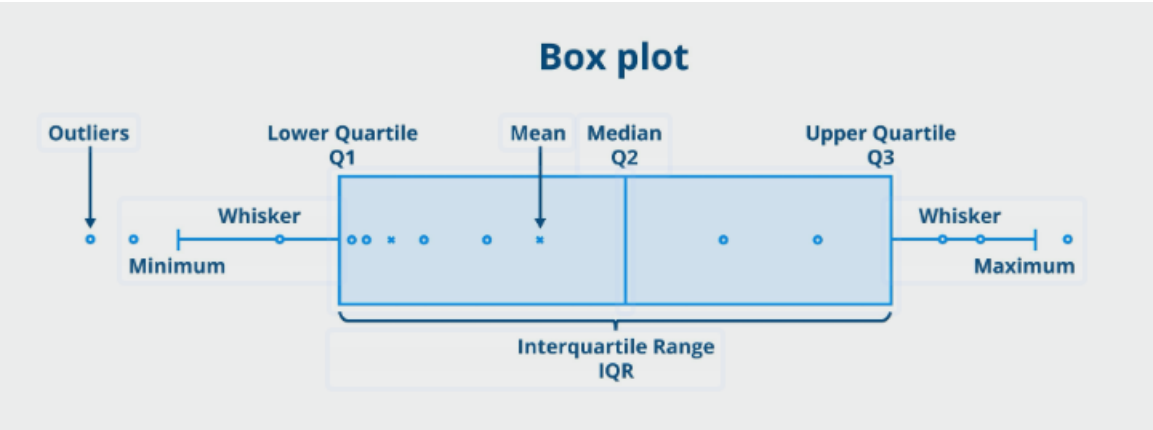
DAY 47 & DAY 48

Day 47: The Interquartile Range (IQR) Strategy

Today we focus on the **IQR Method**, a robust statistical technique for identifying and removing outliers. Unlike the Z-score, which assumes a normal distribution, the IQR method is "distribution-agnostic," making it perfect for skewed data.

1. What is the IQR?

The **Interquartile Range (IQR)** is the distance between the 25th percentile (Q1) and the 75th percentile (Q3). It represents the "middle 50%" of your data.



2. The Mathematical Formula

To find the boundaries for "normal" data, we use the following equations:

1. **Calculate IQR:**

$$IQR = Q_3 - Q_1$$

2. **Define the Lower Bound:**

$$Lower\ Bound = Q_1 - (1.5 \times IQR)$$

3. **Define the Upper Bound:**

$$Upper\ Bound = Q_3 + (1.5 \times IQR)$$

Note: Any data point falling **below** the Lower Bound or **above** the Upper Bound is considered a statistical outlier.

Day 48: Implementing Outlier Removal

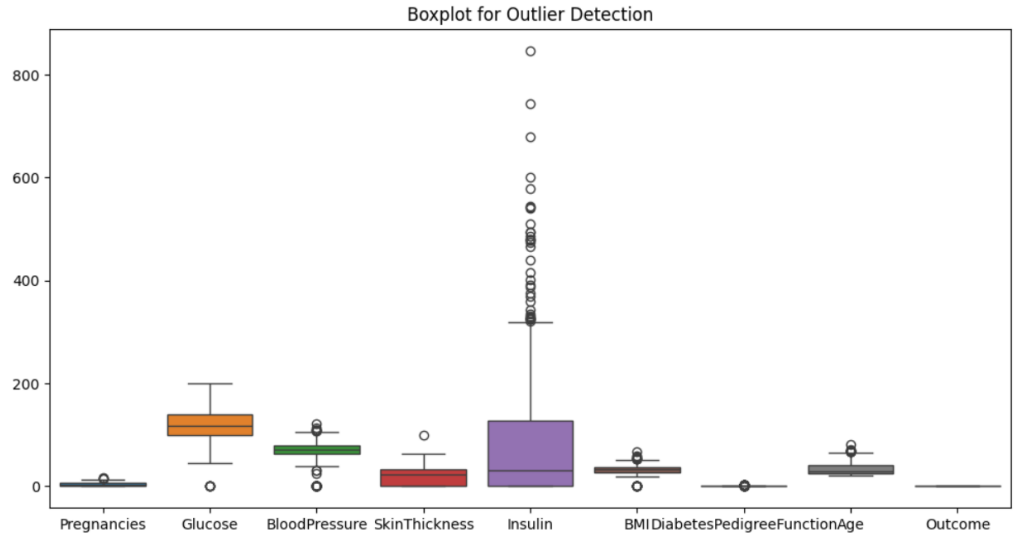
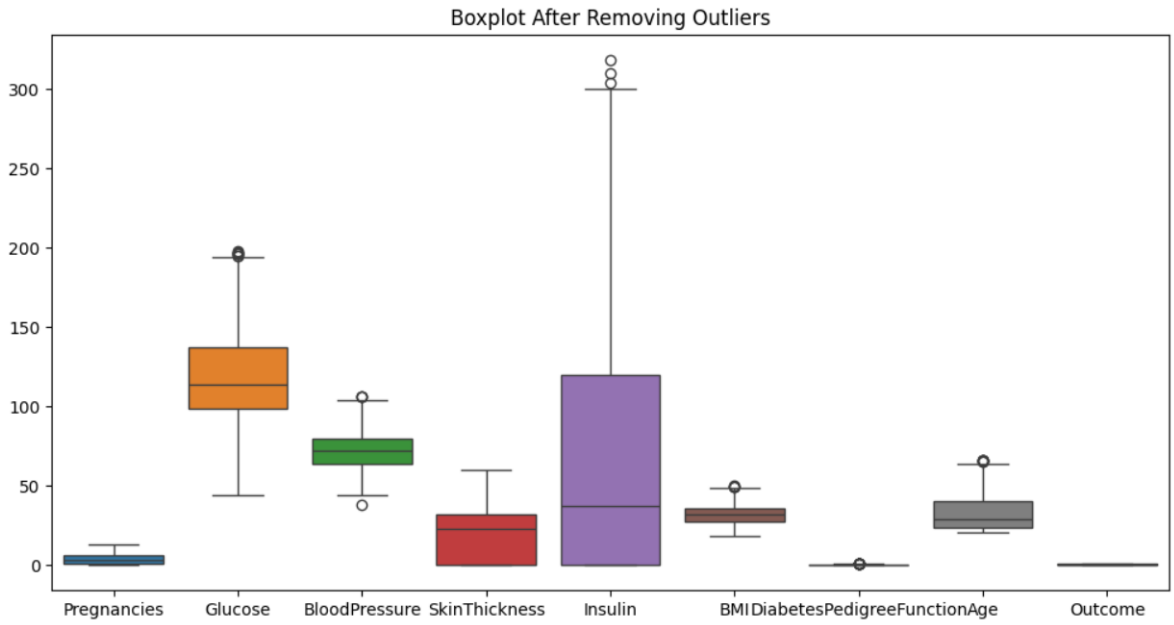
Now that we have the theory, let's put it into practice using Python and the pandas library.

1. The Step-by-Step Workflow

- **Sort the data:** Ensure your values are in ascending order.
- **Calculate Percentiles:** Use .quantile(0.25) and .quantile(0.75).
- **Filter the DataFrame:** Create a new dataset that only includes values within the bounds.

2. Code OUTPUT

BEFORE:



AFTER: